BANKHAUL · PRECIS

BankHaul design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-25	Amish Chadha	Concept A chosen by Amish (O-1, 2026-10-25); station USD 399 with the bank-edge board (O-2)
0.4	2026-10-25	Amish Chadha	Kit 1.8 and 1.9 upgrade: concept options and STEM check linked; STEM fixes D-S1 to D-S7 (stake capacity, stay strength, both loop ends spliced, top pin hole); requirement numbers PR-001 to PR-016
0.3	2026-10-03	Amish Chadha	Far stake driven 3.5 m and R8 restated (Amish's decisions 14A and 15A, BKH-DDR-003); numbers updated
0.2	2026-10-03	Amish Chadha	TRL 2 and 3 update; constructable design (BKH-DDR-001, BKH-DDR-002), first-order numbers, safety
0.1	2026-09-30	Amish Chadha	Initial scaffold

BankHaul design precis

Sets and hauls nets from the bank on a rope loop so fishers stop wading into crocodile water.

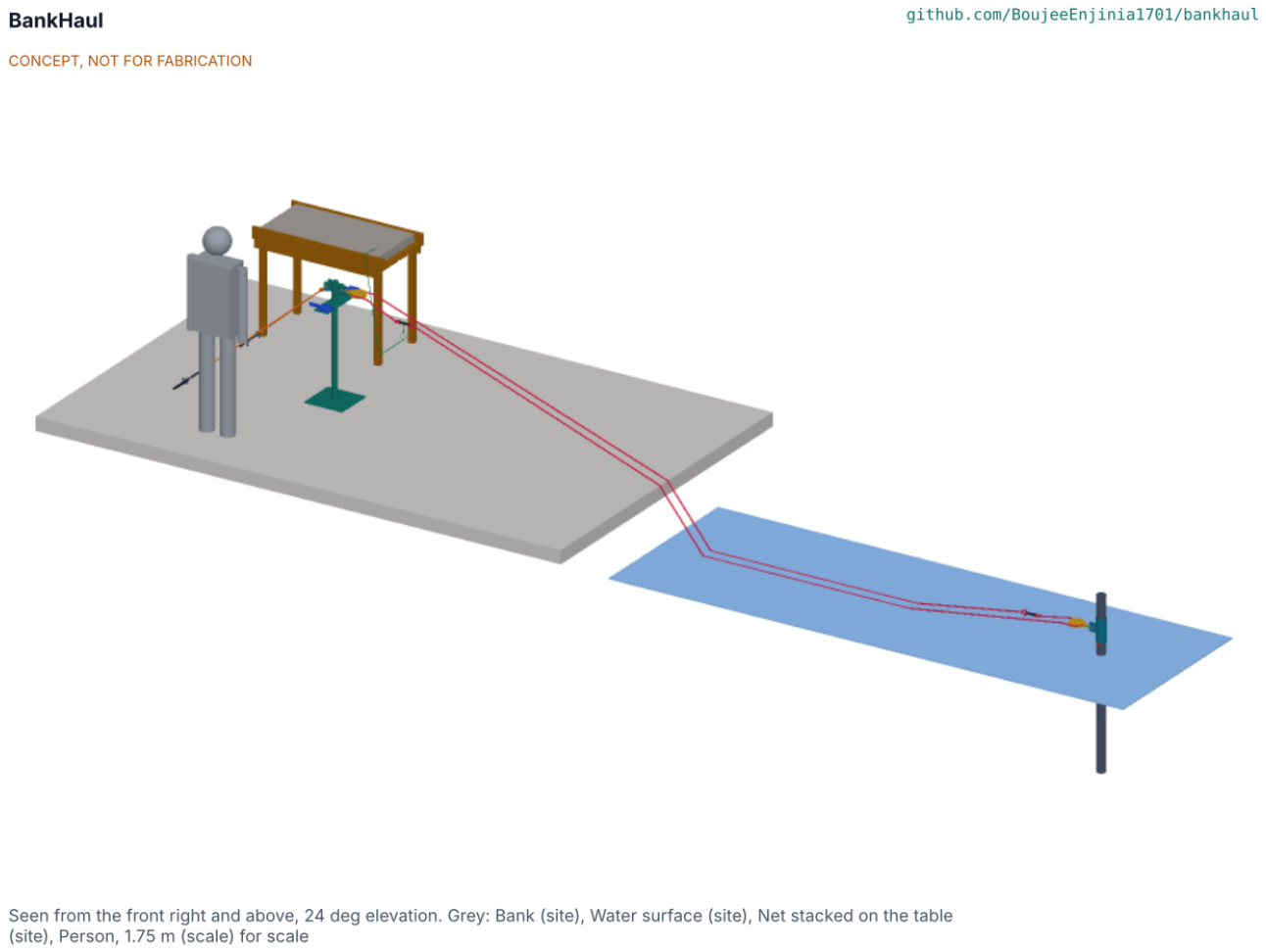


Figure 1. Bank station, rope loop and far stake in the shortened picture layout (7 m between the blocks; 30 m in the design case).

How it works

A rope loop runs from a post on the bank out to a stake in the water and back. Once the stake is set, the fisher never needs to enter the water: the net is clipped to the loop on the bank, pulled out by the loop, and later hauled back onto a table on the bank.

- **Bank station.** A 48.3 mm steel post stands on a 350 mm foot plate with a ground spike, at least 6 m back from the water. Two rope back-stays with turnbuckles hold it back to two screw ground anchors with 250 mm helixes. A swivel block hangs from a pad eye at 950 mm, about waist height. A cleat bar carries two jam cleats, one for each leg of the loop, and a hole to tie off the net's near end.
- **Far end.** A 76.1 mm steel stake, 5.0 m long, is driven 3.5 m into the bed from a boat or at low water. A collar turns on the stake, resting on a through-bolt stop pin that can be moved up or down a hole as the season's water level changes; a second swivel block hangs from the collar, so it lines up with the loop by itself.

- **Loop.** Two halves of 12 mm floating rope, eye-spliced at both ends, are joined by two eye-to-eye swivels. The swivels are the clip rings: when ring 1 is at the bank, ring 2 is at the far block, and each ring stops at a block if pulled too far.
- **Setting.** The fisher clips the far end of the net to ring 1 through a weak link and pulls the other leg of the loop hand over hand; ring 1 carries the net's far end out while the net pays out from the table. When ring 1 reaches the far block, the net lies along the loop. The near end is tied to the cleat bar through a second weak link, and each leg is dropped into its jam cleat.
- **Hauling.** The fisher frees the legs, unties the near end and pulls the net in onto the table hand over hand; the loop follows, bringing ring 1 and the far end in last. The table is at least 5 m from the water.

Components

#	Component	Role	Made or bought
1	Bank post weldment	Carries the near block, the jam cleats and the stays	Made: welded steel tube and plate
2	Far stake	Carries the far end in the bed	Made: steel tube with a welded point and cap
3	Swivel collar	Turns on the stake so the far block lines up with the loop	Made: steel tube and lug
4	Clearing table	Raised surface to stack and clear the net above the waterline	Made: timber
5	Back-stays (2)	Hold the post back against the loop	Made: spliced polyester rope
6	Rope loop	Carries the net out and back	Made: two spliced halves of floating rope
7	Net bridles with weak links (2)	Join each net end to the loop or the post; release at 500 to 700 N	Made: cord, weak link and snap hook
8	Swivel blocks (2)	Turn the loop at each end	Bought: 100 mm sheave for 12 mm rope
9	Swivel rings (2)	Join the loop halves; clip rings and end stops	Bought
10	Jam cleats (2)	Hold each leg; the bank brake	Bought
11	Screw ground anchors (2)	Anchor the stays in soft soil	Bought: 250 mm helix, 1.6 m rod
12	Turnbuckles (2)	Tension the stays	Bought: rated M12
13	Shackles (6)	Join blocks, stays and anchors	Bought: rated 10 mm
14	Stop pin	Holds the collar at the season's level	Bought: M12 bolt

Concept options and STEM check

This design is option A of three concept options scored against the user requirements (BKH-OPT-001): A, the running rope loop, scores 178 of 220, C, a counterweighted shore lift net, 149 and B, an overhead cableway, 139; Amish chose A on 2026-10-25 (BKH-DDR-004). The STEM check (BKH-STM-001) finds the loop within known science after seven detail fixes made under Amish's standing instruction of 2026-10-23 (BKH-DEC-001 D-S1 to D-S7); it stays open on field data.

Key design choices

Each choice is recorded in BKH-DDR-001 (TRL 2 review) or BKH-DDR-002 (design for construction) and listed in `docs/06-design-decisions.md`.

- **A stayed post, not a cantilever.** A post holding 3 kN at waist height as a cantilever would need a deep foundation; stays to screw anchors take the pull as tension and leave the post in compression, and two people can move it.
- **Rings that never pass a block.** The loop's joins are the clip rings, placed half a loop apart, so with a net shorter than the span less 2 m no ring or knot ever runs round a sheave.
- **Weak links.** Every net end reaches the loop or post through a link that releases at 500 to 700 N, about eight times the largest load in normal use, so a crocodile or snag taking the net cannot load the far stake past 1.4 kN or pull a person.
- **Floating loop.** Polysteel floats, so the loop stays visible, clear of the bed and easy to reach from a boat.
- **Hand pull only.** No winch or capstan (IP screen design-around); the peak pull of about 200 N does not need one.

First-order numbers

All figures from BKH-CAL-001; assumptions are stated there.

Quantity	Value	Basis
Loop length, 30 m span	60.4 m	Two legs plus half a wrap round each 112 mm rope circle
Peak hand pull to haul a 30 m net with 10 kg catch	about 200 N	Catch, weed and a fifth of the wet net up a 27 degree bank edge
Pull to set the net	about 43 N	Loop and net paying out at 0.5 m/s
Weak link release	500 to 700 N	Nominal 600 N, plus or minus 15 %
Far end design load	1.4 kN	Both legs at the link's highest release
Stay tension at the 3 kN bank load	2.1 kN each, factor 6.4 on 10 mm polyester	Stays 39 degrees up, 22 degrees either side
Screw anchor capacity	4.4 kN each	250 mm helix, 0.9 m deep, soil of 10 kPa shear strength
Far stake capacity	4.2 kN in soft mud (the stake yields first); 3.0 kN in very soft mud (factor 2.03 on PR-005)	Broms, the lesser of the short and long free-head pile, 3.5 m embedded (D-S2)
Cycle time	about 6 minutes	Without clearing fish
Station mass	about 106 kg; heaviest part 33.2 kg (far stake)	Model volumes and catalogue figures
Station parts	USD 399 (steel trial station; R8 now applies to a local-materials station costed with the co-design partner)	`bom/bom.csv`

Value-engineering target: USD 1,800. Estimated cost of the constructable design: USD 409 including the stake driving cap (USD 1,391 under the target).

Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice):

- No powered hauler: hand-pulled loop only, which keeps clear of motorised line-hauler patents such as EP1429601 (expired) and later filings.
- Commodity blocks, anchors and rope, following the traditional running-line practice, rather than any proprietary downrigger or hauler pulley.

Shared blocks

- CalRig proof-load: the first candidate rig for the anchor, stake and weak link tests at TRL 4.
- Hand capstan block from SaltDrag and SiltHaul: not used; the peak pull is within one person's reach, and the design-around excludes powered hauling.

Safety

> **Safety:** BankHaul reduces time in the water; it does not protect anyone from a crocodile that leaves the water. Keep away from the water's edge, especially at night and at dawn, and follow the local wildlife authority's advice.
>
> Placing the far stake is the most dangerous step: do it from a boat or at low water with a second person watching, never by wading.
>
> A loaded rope can snap back. Never stand in the bight of the loop or in line with a loaded stay; check anchors, stays, blocks and rope before each season and after any hard pull.
>
> Never tie the loop or the net to your body, and never wrap the rope round a hand. If something big takes the net, let go: the weak link and the jam cleats hold the rest.
>
> Take the loop out of the water before a flood; floating debris can load it far beyond its design.
>
> This design is published as an open engineering reference. It is not certified equipment.

Open questions

Open decisions and items to confirm when parts are bought are kept in the design decisions register (`docs/06-design-decisions.md`).